



Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no.....

Name of the Course: B.Tech (CSE)-AI & ML
Semester: 4th
Name of the Paper: Programming in Java
Paper Code: TCS-408
Time: 1.5 hours

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

Q1. (10 Marks)

- a. Discuss jagged array with a proper code. Also, discuss autoboxing and unboxing with the help of examples. (CO1)

OR

- b. Discuss any eight features of Java. Also, provide the code for any four features. (CO1)

Q2. (10 Marks)

- a. Discuss the different types of constructors with suitable codes. Also write a program for the following scenario.

A bank wants to store balances of 5 customers, calculate the total balance and display all balances greater than a threshold (e.g., INR 50,000). (CO1)

OR

- b. Discuss the internal architecture of Java Virtual Machine with the help of a neat and clean diagram. (CO1)

Q3. (10 Marks)

- a. Discuss the importance of packages in Java. Create a package with a class *Student* with instance variables *id*, *name* and instance methods *void setDetails(int id, String name)* and *void showDetails()*. Create another class *MainClass* outside the package which uses the *Student* class. From this *MainClass* set and show all the details of any student. (CO2)

OR

- b. Explain the advantages of inheritance and list the types of inheritance with suitable codes. Explain the reasons behind the statement that "Java does not support multiple inheritance through classes". How does Java overcome this problem using interface? (CO2)

Q4. (10 Marks)

- a. Discuss the different ways to import Java packages with suitable codes. Also discuss *public static void main(String args[])* in detail. (CO1 & CO2)

OR

- b. Explain *this*, *final* and *super* keywords with suitable codes. Also write a program to show constructor chaining. (CO2)

Q5. (10 Marks)

- a. Discuss the access scope of public, protected, default and private access specifiers with suitable examples. Also explain abstract class and interface in detail. (CO2)

OR

- b. Write a Java program to solve the following scenario using inheritance:
There is a class *Vehicle* with data members: *String brand*, *int speed* and methods as:
void setDetails(String brand, int speed): to set brand name and speed
void showDetails(): to show the brand name and speed



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Create a child class *Car* that inherits *Vehicle* and has data member: *int doors* (for number of doors) and methods as:

void setDoors(int doors): to set the number of doors.

void displayCarDetails() to display complete details (brand, speed and number of doors).

Write a *MainClass* to run the program.

(CO2)